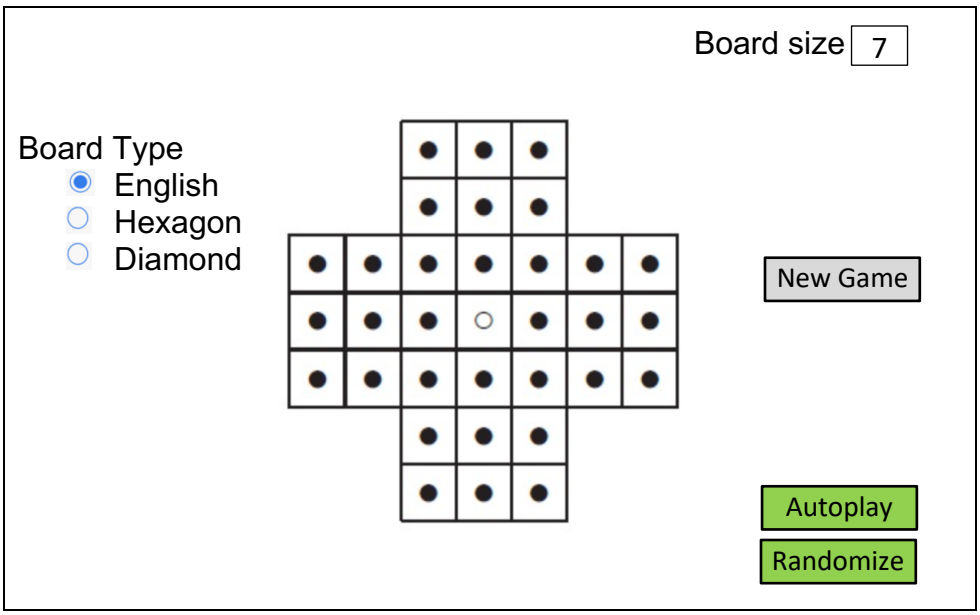
Project Sprint #3

The Solitaire game is described in CS449HomeworkOverview.docx. You should read the description very carefully.

Your submission must include the GitHub link to your project and you must ensure that the instructor has the proper access to your project. You will receive no points otherwise.

GitHub link:

Implement all the features that support a human player to play a manual or automated Solitaire game against a human opponent and refactor your existing code if necessary. The minimum features include **choosing the game mode (manual or automated), choosing the board size, setting up a new game, making a move (in a manual or automated game), randomize the state of the board** and **determining if a manual or automated game is over**. The following is a sample GUI layout. It is required to use a class hierarchy to deal with the common requirements of the Manual Game and the Automated Game. **If your code for Sprint 2 has not considered class hierarchy, it is time to refactor your code**.



Deliverables: expand and improve your submission for sprint 2.

1. **Demonstration (9 points)**
   Submit a video of no more than five minutes, clearly demonstrating the following features.
   (a) A manual game until the game is over for English board
   (b) A manual game until the game is over for Hexagon board
   (c) An automated game until the game is over
   (d) A manual game with periodic randomization of states until the game is over
   (e) Some automated unit tests for the manual game mode
   (f) Some automated unit tests for the automated game mode

   In the video, you must explain what is being demonstrated.

2. Summary of Source Code (1 points)

Source code file name	Production code or test code?	# lines of code
	Total	

You must submit all source code via github to get any credit for this assignment.

3. Production Code vs User stories/Acceptance Criteria (3 points)

Summarize how each of the user story/acceptance criteria is implemented in your production code (class name and method name etc.)

User Story ID	User Story Name
1	Choose a board size and type
2	Choose the game mode of a chosen board (manual/automated)
3	Start a new game of the chosen board size and game mode
4	Make a move in a manual game
5	A manual game is over
6	Make a move in an automated game
7	An automated game is over
8	Randomize the state of the board during a manual game

User Story ID	AC ID	Class Name(s)	Method Name(s)	Status (complete or not)	Notes (optional)
1	1.1				
	1.2				
	…				
2	2.1				
	…				

4. Tests vs User stories/Acceptance Criteria (2 points)

Summarize how each of the user story/acceptance criteria is tested by your test code (class name and method name) or manually performed tests.

User Story ID	User Story Name
1	Choose a board size and type
2	Choose the game mode of a chosen board (manual/automated)
3	Start a new game of the chosen board size and game mode
4	Make a move in a manual game
5	A manual game is over
6	Make a move in an automated game
7	An automated game is over
8	Randomize the state of the board during a manual game

4.1 Automated tests directly corresponding to some acceptance criteria

User Story ID	Acceptance Criterion ID	Class Name (s) of the Test Code	Method Name(s) of the Test Code	Status (Pass/Fail/In Progress)
1	1.1			
	1.2			
	…			
2	2.1			
	…			

4.2 Manual tests directly corresponding to some acceptance criteria (if any)

User Story ID	Acceptance Criterion ID	Test Case Input	Test Oracle (Expected Output)	Notes
1	1.1			
	1.2			
	…			
2	2.1			
	…			

4.3 Other automated or manual tests not corresponding to the acceptance criteria (if any)

Number	Test Input	Expected Result	Class Name of the Test Code	Method Name of the Test Code

5. **(Part 1) Describe how the class hierarchy in your design deals with the common and different requirements of the Manual Game and the Automated Game.**
   At least 1/2 page, excluding screenshots/diagrams (12pt, single-spaced), is required. (3 Points)

   (Part 2) Demonstrate how you use LLM (ChatGPT or other) to analyze how well your code adheres to the design principles discussed in class - modularity, cohesion, coupling, and encapsulation – using the definitions provided in class. Provide screenshots of your interactions with the LLM, showing the prompts used and the responses received.
   Interpret the LLM's feedback, refine your prompts if needed to obtain more relevant responses, and discuss any changes made to your code based on the analysis. Explain how these refinements improve adherence to the design principles.
   At least 1/2 page is required, excluding screenshots (12pt, single-spaced). (2 Points)